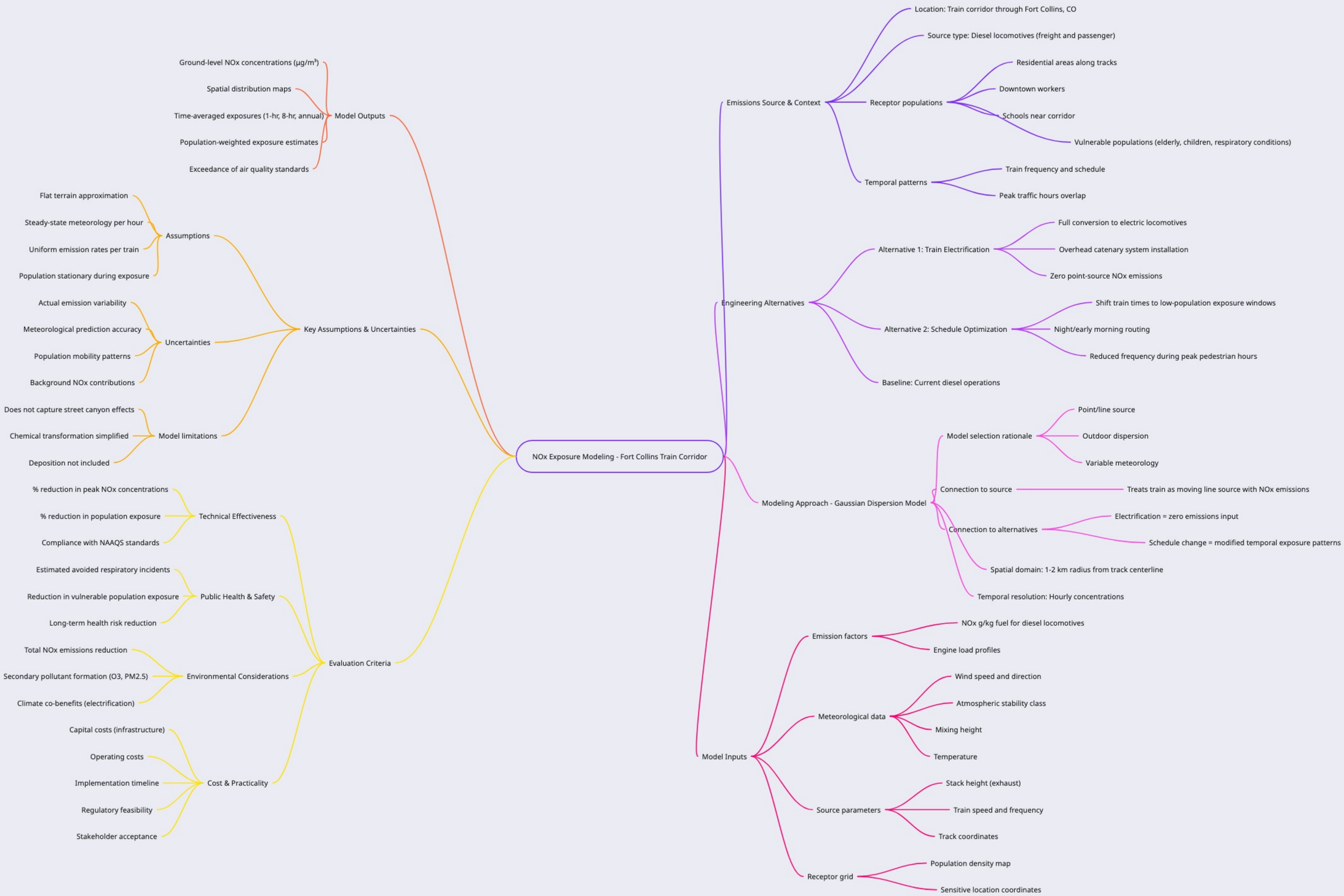
NOx Exposure Modeling - Fort Collins Train Corridor
Emissions Source & Context
Location: Train corridor through Fort Collins, CO
Source type: Diesel locomotives (freight and passenger)
Receptor populations
Residential areas along tracks
Downtown workers
Schools near corridor
Vulnerable populations (elderly, children, respiratory conditions)
Temporal patterns
Train frequency and schedule
Peak traffic hours overlap
Engineering Alternatives
Alternative 1: Train Electrification
Full conversion to electric locomotives
Overhead catenary system installation
Zero point-source NOx emissions
Alternative 2: Schedule Optimization
Shift train times to low-population exposure windows
Night/early morning routing
Reduced frequency during peak pedestrian hours
Baseline: Current diesel operations
Modeling Approach - Gaussian Dispersion Model
Model selection rationale
Point/line source
Outdoor dispersion
Variable meteorology
Connection to source
Treats train as moving line source with NOx emissions
Connection to alternatives
Electrification = zero emissions input
Schedule change = modified temporal exposure patterns
Spatial domain: 1-2 km radius from track centerline
Temporal resolution: Hourly concentrations
Model Inputs
Emission factors
NOx g/kg fuel for diesel locomotives
Engine load profiles
Meteorological data
Wind speed and direction
Atmospheric stability class
Mixing height
Temperature
Source parameters
Stack height (exhaust)
Train speed and frequency
Track coordinates
Receptor grid
Population density map
Sensitive location coordinates
Model Outputs
Ground-level NOx concentrations (μg/m³)
Spatial distribution maps
Time-averaged exposures (1-hr, 8-hr, annual)
Population-weighted exposure estimates
Exceedance of air quality standards
Key Assumptions & Uncertainties
Assumptions
Flat terrain approximation
Steady-state meteorology per hour
Uniform emission rates per train
Population stationary during exposure
Uncertainties
Actual emission variability
Meteorological prediction accuracy
Population mobility patterns
Background NOx contributions
Model limitations
Does not capture street canyon effects
Chemical transformation simplified
Deposition not included
Evaluation Criteria
Technical Effectiveness
% reduction in peak NOx concentrations
% reduction in population exposure
Compliance with NAAQS standards
Public Health & Safety
Estimated avoided respiratory incidents
Reduction in vulnerable population exposure
Long-term health risk reduction
Environmental Considerations
Total NOx emissions reduction
Secondary pollutant formation (O3, PM2.5)
Climate co-benefits (electrification)
Cost & Practicality
Capital costs (infrastructure)
Operating costs
Implementation timeline
Regulatory feasibility
Stakeholder acceptance